

Drawing on a canvas

To create graphics using Python, you need to make a blank area to draw on. This is known as a canvas. You can use x and y coordinates to tell Python exactly where to draw on the canvas.

1 Create a graphics program

Use this code to create a window and put a canvas inside it. It will then draw random circles on the canvas.

```
from random import *
from tkinter import *
size = 500
window = Tk()
canvas = Canvas(window, width=size, height=size)
canvas.pack()
while True:
    col = choice(['pink', 'orange', 'purple', 'yellow'])
    x0 = randint(0, size)
    y0 = randint(0, size)
    d = randint(0, size/5)
    canvas.create_oval(x0, y0, x0 + d, y0 + d, fill=col)
    window.update()
```

This imports the "randint" and "choice" functions from the Random module

This imports all of the Tkinter functions

The variable "size" sets the dimensions of the canvas

This creates a canvas inside a window

A forever loop makes the program draw circles endlessly

This chooses a random color from the list

This creates a circle of a random size in a random place on the canvas

This part of the line draws the circle

This part fills it with the color that has been chosen ("col")

2 Colored canvas

Run the code and the program will start drawing circles on the canvas.

Circles are drawn in random places



The size of each circle is random

EXPERT TIPS

Coordinates

In Tkinter, x coordinates get larger moving to the right, and y coordinates get larger moving downward. (0,0) is in the top-left corner.

